

AI Study Assistant Agent

An Automated Educational Chatbot Built with n8n & Groq

Domain: Artificial Intelligence / EdTech

Orchestration: n8n Workflow Automation

Model Provider: Groq Cloud API

1. PROJECT OVERVIEW

The **AI Study Assistant Agent** is an interactive workflow-based educational assistant developed using **n8n** node-based automation platform. The system is designed to provide immediate academic guidance to engineering students studying Computer Science subjects. By integrating LLM capabilities via **Groq**, the agent can understand technical queries, explain core computing concepts, generate exam-ready short answers, and deliver practical code snippets.

2. OBJECTIVE

The primary objectives of this project are:

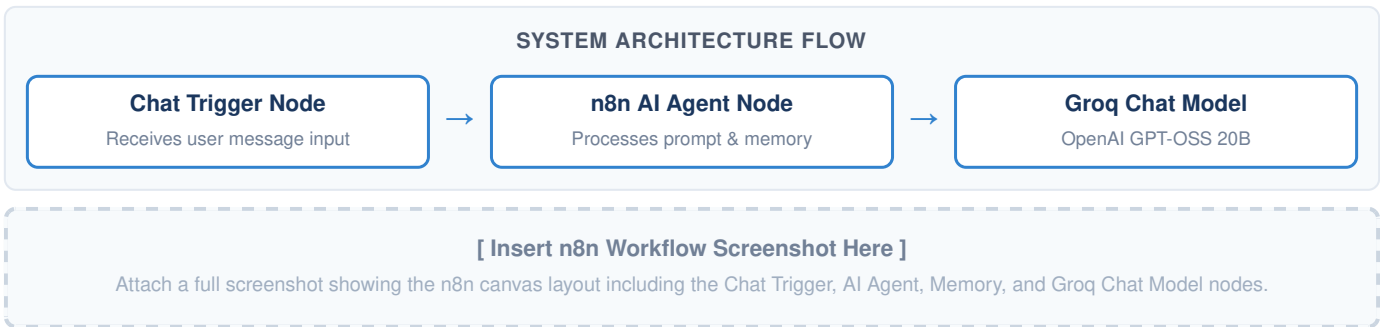
- To automate student inquiry handling using a structured, workflow-driven AI agent.
- To provide accurate, fast, and structured academic support in fundamental Computer Science subjects including Java, Python, C, DBMS, Data Structures & Algorithms (DSA), HTML/CSS, and foundational AI concepts.
- To deliver answers in flexible styles—ranging from simple English to a natural Hindi-mixed context (Hinglish) based on student convenience.
- To maintain context across ongoing conversations using structured chat memory nodes.

3. TECHNOLOGIES USED

Component	Technology / Model	Role in System
Workflow Engine	n8n (Node-based Orchestration)	Handles incoming chat events, manages node flow, state, and API routing.
AI Model Provider	Groq API	Provides high-speed LLM inference hardware interface.
LLM Backbone	OpenAI GPT-OSS 20B	Generates intelligent, subject-focused responses based on system prompt.
State Memory	n8n Conversation Memory Node	Stores conversational context for multi-turn chat sessions.

4. SYSTEM WORKFLOW

The system follows an event-driven flow triggered whenever a user submits a chat message. The conversation flow passes through three core stages as shown below:



5. WORKING OF THE AI AGENT

The core logic of the project resides in the **n8n AI Agent Node**, which acts as the decision coordinator between user input, system memory, and the Groq LLM endpoint.

System Instructions & Rules

1. Subject Scope: Restrict answers strictly to Java, Python, C, DBMS, DSA, HTML/CSS, and AI concepts.
 2. Style & Tone: Explain concepts simply. Provide short, structured, exam-oriented points when requested.
 3. Code & Examples: Provide minimal, accurate code snippets and practical examples when relevant.
 4. Support: Support simple English as well as Hindi-mixed language for clear student understanding.
 5. Reliability: Ask for clarification if a query is ambiguous. Do not invent or fabricate facts.
 6. Behavioral Stance: Maintain a polite, helpful, clear, and concise tone.

6. KEY FEATURES

- **Curriculum-Aligned Assistance:** Focuses specifically on key engineering CS topics.
- **Exam-Oriented Explanations:** Produces concise bullet points ideal for rapid revision and university assessments.
- **Multilingual / Hindi-Mixed Support:** Adapts to informal student communication for better concept clarity.
- **Contextual Memory:** Remembers previous user messages within the session to handle follow-up queries seamlessly.
- **Low Latency Response:** Powered by Groq's fast inference Engine for near-instant responses.

7. SAMPLE QUERIES AND RESPONSES

User: "What is DBMS?"

AI Study Assistant:
A Database Management System (DBMS) is software that lets you create, manage, and manipulate databases. Think of it as the bridge between users (or applications) and the raw data stored on disk.

Key Features:

- **Data Storage:** Keeps data in files on disk/SSD, organized for fast access. Ensures durability and recovery.
- **Data Retrieval & Manipulation:** Provides query languages (SQL/NoSQL) and supports insert, update, delete operations.
- **Concurrency Control & Security:** Manages simultaneous user access, handles authentication, authorization, and encryption.
- **Transaction Management & ACID:** Guarantees reliable, predictable changes via ACID properties.

Logical vs Physical: Logical schema designs tables/relationships, while physical storage translates it into disk files and indexes.

User: "Explain inheritance in Java."

AI Study Assistant:
Inheritance lets one class (the subclass) acquire the fields and methods of another class (the superclass). It expresses an "is-a" relationship (e.g., Dog is an Animal). Java supports single-inheritance for classes using the extends keyword.

Code Example:

```
public class Animal {
    protected String name;
    public Animal(String name) { this.name = name; }
    public void speak() { System.out.println(name + " makes a noise."); }
}
public class Dog extends Animal {
    public Dog(String name) { super(name); }
    @Override public void speak() { System.out.println(name + " says: Woof!"); }
}
```

Key Rules: Subclasses inherit public/protected members; private members are not directly accessible. The super keyword is used to call superclass constructors/methods.

[Insert Chat Demo Screenshot Here]

Attach a screenshot of the interactive chat session showing sample student interactions.

8. TESTING & VALIDATION

The workflow was systematically tested within the n8n canvas using real-time user chat inputs. Key test scenarios evaluated include:

- **Basic Conceptual Questions:** Tested with queries like “*What is DBMS?*” to verify clarity and structural formatting.
- **Object-Oriented Programming Concepts:** Tested with queries like “*Explain inheritance in Java.*” to check code sample accuracy and technical precision.
- **Out-of-Scope Queries:** Verified that the agent politely redirects or handles non-computer science queries while remaining within defined bounds.
- **Memory Recall:** Executed multi-turn interactions (e.g., asking for an example after defining a concept) to verify conversation continuity.

9. FUTURE SCOPE

- **Vector DB Integration:** Connecting external vector stores (e.g., Qdrant, Pinecone) to allow RAG-based query answering over uploaded university textbooks and PDF lecture notes.
- **Multi-Channel Interface:** Deploying the n8n chat workflow across communication tools such as Telegram, Discord, or WhatsApp.
- **Code Execution Sandbox:** Adding an automated tool-calling node to test user-submitted Python or C code in a secure execution environment.

10. CONCLUSION

The **AI Study Assistant Agent** successfully demonstrates how low-code workflow automation via n8n can be combined with high-performance LLM infrastructure provided by Groq. The resulting solution provides a practical, reliable, and responsive educational assistant tailored specifically for engineering students pursuing Computer Science coursework.